

Wick's theorem

W. Ryssens



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...learned how to write operators with them

A few more words on operators

Operators

Operators in first quantisation

$$\text{1-body } F = \sum_{i=1}^N f(i).$$

$$\text{2-body } G = \frac{1}{2} \sum_{i \neq j=1}^N g(i, j).$$

Sums over particles
operators defined on $\mathcal{H}_N$

Operators in **second** quantisation

$$\text{1-body } F = \sum_{\alpha\beta} f_{\alpha\beta} a_{\alpha}^{\dagger} a_{\beta}.$$

$$\text{2-body } G = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} g_{\alpha\beta\gamma\delta} a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}.$$

Sums over states in $\mathcal{H}_1$
operators defined on $\mathcal{F}$!

Ordering!

Operator matrix elements

With (iso)spin omitted:

$$f_{\alpha\beta} = \langle 1 : \alpha | f(1) | 1 : \beta \rangle = \int d\vec{r}_1 d\vec{r}'_1 \psi_{\alpha}^*(\vec{r}_1) f(\vec{r}_1, \vec{r}'_1) \psi_{\beta}(\vec{r}'_1).$$

Examples: external potentials, kinetic energy, angular momentum, magnetic moments ...

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Examples: external potentials, kinetic energy, angular momentum, magnetic moments ...

$$\begin{aligned} g_{\alpha\beta\gamma\delta} &= \langle 1 : \alpha, 2 : \beta | g(1, 2) | 1 : \gamma, 2 : \delta \rangle \\ &= \int d\vec{r}_1 d\vec{r}'_1 d\vec{r}_2 d\vec{r}'_2 \psi_{\alpha}^*(\vec{r}_1) \psi_{\beta}^*(\vec{r}_2) g(\vec{r}_1, \vec{r}_2, \vec{r}'_1, \vec{r}'_2) \psi_{\gamma}(\vec{r}'_1) \psi_{\delta}(\vec{r}'_2). \end{aligned}$$

Examples: gravity, Coulomb interaction, all pairwise interactions, ...

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These operators are often "local" = diagonal in position space

$$f(\vec{r}_1, \vec{r}'_1) \sim \delta(\vec{r}_1 - \vec{r}'_1) \text{ or } g(\vec{r}_1, \vec{r}_2, \vec{r}'_1, \vec{r}'_2) \sim \delta(\vec{r}_1 - \vec{r}'_1) \delta(\vec{r}_2 - \vec{r}'_2)$$

Examples: single-particle hamiltonian

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such that

$$\hat{h} = \sum_{\alpha} \epsilon_{\alpha} \hat{a}_{\alpha}^{\dagger} \hat{a}_{\alpha}.$$

counts the number of particles.

Examples: the particle number operator $\hat{N}$

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Q: why?

Q for next time: can you find the first quantised form by reverse engineering?

Q for next time: demonstrate that $|\alpha\beta\rangle$ is an eigenstate of $\hat{N}$. What is its eigenvalue?

The nuclear Hamiltonian

Note: we forget about 3-body forces today!

$$\hat{H} = \sum_{i=1}^N \frac{\hat{p}_i^2}{2m} + \frac{1}{2} \sum_{i \neq j=1}^N \hat{V}(i,j)$$

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You will learn more about this in exercice session 1!

The one-body density matrix

The **one-body density matrix** ρ of a Slater determinant

$$\rho_{\mu\nu} = \langle \alpha\beta \dots | \hat{a}_\nu^\dagger \hat{a}_\mu | \alpha\beta \dots \rangle .$$

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or in coordinate-spin-isospin-space

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Q: what do you get when summing over σ, τ and setting $\vec{r} = \vec{r}'$?

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If $|\mu\rangle$ and $|\nu\rangle$ are in the basis underpinning the Slater determinant, then

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The one-body density ρ will not take this form in an arbitrary basis. What is fundamental is that it is a **projector**, i.e.

$$\rho^2 = \rho .$$

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Today

All expectation of Slater determinants can be computed from 'just' ρ .

Wick's theorem

The set-up

- 1 A reference state $|\Psi\rangle$
 $|\Psi\rangle$ does not have to be a Slater determinant!
- 2 A **complete** set of $\hat{c}_\alpha, \hat{c}_\alpha^\dagger$ such that

$$\hat{c}_\alpha |\Psi\rangle = 0, \quad \{c_\alpha, c_\beta^\dagger\} = \delta_{\alpha\beta}.$$

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Trivial example: $|\Psi\rangle \equiv |0\rangle$ with the standard ladder operators $\hat{a}_i^\dagger, \hat{a}_i$.

Slater determinant as vacuum

Is $|\Psi\rangle \equiv |\alpha\beta\dots\rangle$ a vacuum for the standard ladder operators $\hat{a}_\mu, \hat{a}_\mu^\dagger$?

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It is however a vacuum for the following choice:

$$\left\{ \begin{array}{ll} \hat{\beta}_\alpha \equiv \hat{a}_\alpha^\dagger & \text{if } \alpha \text{ is occupied.} \\ \hat{\beta}_\alpha \equiv \hat{a}_\alpha & \text{if } \alpha \text{ is empty.} \end{array} \right.$$

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'Fermi vacuum'

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- 1 Consider $|\Phi\rangle = \prod_{i=1}^N \hat{a}_i^\dagger |0\rangle$, vacuum w.r.t. qp operators $\hat{\beta}_i$

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Later on we will see this kind of transformation as a 'trivial' one!

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Consider

- $|\Phi\rangle$, a reference vacuum for the complete set $\hat{c}_\alpha$
- some operator $\hat{A}$ = a string of annihilation and creation operators

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you can further permute
creation or annihilation operators.

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Reminder

The normal product is defined with respect to a reference vacuum $| \Phi \rangle$!

Normal ordering: basis transformation

What if $\hat{A} \equiv \hat{a}_\alpha \hat{a}_\beta^\dagger \hat{a}_\gamma^\dagger \hat{a}_\delta$ is expressed in terms of $\hat{a}_\mu$...

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Q for next time: repeat this for $\hat{A} = \hat{a}_\alpha \hat{a}_\beta^\dagger$ and $\hat{a}_\alpha = \sum_\lambda U_{\alpha\lambda} \hat{\beta}_\lambda + V_{\alpha\lambda}^* \hat{\beta}_\lambda^\dagger$.

Elementary contractions

The contraction of a product $\hat{A}\hat{B}$ of two operators $\hat{A}$ and $\hat{B}$ is defined as

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...because $AB - :AB:$ can be expressed in terms of anticommutators

Elementary contractions

The most important property

The contraction is a number anyway!

$$\begin{aligned}\overline{\hat{A}\hat{B}} &= \langle \Phi | \hat{A}\hat{B} | \Phi \rangle \\ &= \langle \Phi | \hat{A}\hat{B} | \Phi \rangle - \langle \Phi | : \hat{A}\hat{B} : | \Phi \rangle \\ &= \langle \Phi | \hat{A}\hat{B} | \Phi \rangle.\end{aligned}$$

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The contraction is the expectation value w.r.t. the vacuum!

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Notes

- Only one single contraction is non-trivial.
- The operator products were normal-ordered already in three cases.

Elementary contractions

Example 2: the Fermi vacuum $|\Phi\rangle = \prod_{\alpha=1}^N \hat{a}_{\alpha}|0\rangle$
with quasi-particle operators

$$\begin{cases} \hat{\beta}_{\alpha} = \hat{a}_{\alpha}^{\dagger} & \text{for } \alpha = 1, \dots, N & \rightarrow \text{hole operators} \\ \hat{\beta}_{\alpha} = \hat{a}_{\alpha} & \text{for } \alpha = N+1, \dots & \rightarrow \text{particle operators} \end{cases}$$

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...when $|\Phi\rangle$ is a Slater determinant!

Elementary contractions

Realistic example:

- Fermi vacuum $|\Phi\rangle$
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- particle operators $\hat{a}$.
- operators $\hat{A}$ and $\hat{B}$ are expressed in a different basis! $\hat{b}_\mu^\dagger = \sum_\alpha C_{\alpha\mu} a_\alpha^\dagger$

Density matrix in the
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Q for next time: prove these!

Wick's theorem: why?

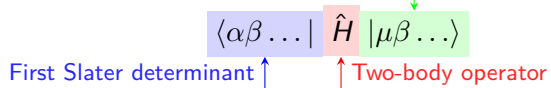
Imagine you want to evaluate a matrix element that looks like this

Slater determinant with α emptied and μ filled

$$\langle \alpha \beta \dots | \hat{H} | \mu \beta \dots \rangle$$

First Slater determinant

Two-body operator

The diagram shows the matrix element $\langle \alpha \beta \dots | \hat{H} | \mu \beta \dots \rangle$. The bra part $\langle \alpha \beta \dots |$ is enclosed in a light blue box, and the ket part $| \mu \beta \dots \rangle$ is enclosed in a light green box. A light red box contains the operator $\hat{H}$. A green arrow points from the text 'Slater determinant with α emptied and μ filled' to the green box. A blue arrow points from the text 'First Slater determinant' to the blue box. A red arrow points from the text 'Two-body operator' to the red box.

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In first quantisation, the steps would be

- 1 Expand $\langle \alpha \beta \dots |$ and $|\mu \beta \dots \rangle$ in terms of product states.
- 2 Calculate $\langle 1 : p(1), \dots | \hat{H} | 1 : p_2(1), \dots \rangle$ for all permutations p_1, p_2 .

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But there are $(N!)^2$ of these matrix elements ...:(

Wick's theorem: why?

Imagine you want to evaluate a matrix element that looks like this

Slater determinant with α emptied and μ filled

$$\langle \alpha \beta \dots | \hat{H} | \mu \beta \dots \rangle$$

First Slater determinant Two-body operator

Wick's theorem: why?

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In **second quantisation**, the steps are

- 1 Take $|\alpha \beta \dots\rangle = |\Phi\rangle$ as a reference Fermi vacuum
- 2 Express $|\mu \beta \dots\rangle = \hat{a}_\mu^\dagger \hat{a}_\alpha |\Phi\rangle = |\Phi_\alpha^\mu\rangle$, a **particle-hole** excitation.
- 3 Write down $\hat{H}$ in the basis associated with Φ , i.e. $\mathcal{B}_1 = \{\hat{a}_\alpha\}$.

Wick's theorem: why?

$$\langle \alpha \beta \dots | \hat{H} | \mu \beta \dots \rangle = \frac{1}{2} \sum_{\beta \gamma \delta \epsilon} v_{\beta \gamma \delta \epsilon} \langle \Phi | \hat{a}_{\beta}^{\dagger} \hat{a}_{\gamma}^{\dagger} \hat{a}_{\epsilon} \hat{a}_{\delta} \hat{a}_{\mu}^{\dagger} \hat{a}_{\alpha} | \Phi \rangle$$

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...or quasi-particle creation operators to the left.

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- ✓ Use Gian Carlo Wick's theorem!

Wick's theorem: why?

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Doable, but **tedious**.

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...which is directly exploiting the pattern of the tedious calculations.

Wick's theorem

The product of n **creation or annihilation operators** is equal to the sum of all the terms obtained by

- contracting p pairs of operators ($0 \leq p \leq \frac{n}{2}$)
- multiplying by the normal product of the uncontracted operators
- multiplying by the signature of the permutation used to bring contracted operators next to each other

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all of these can be creation/annihilation operators

$$\begin{aligned}
 ABCD \dots YZ &= \overbrace{ABCD \dots YZ}^{p=0} - \overbrace{ACBD \dots YZ}^{p=1} + \overbrace{ADBC \dots YZ}^{p=1} - \overbrace{ACDB \dots YZ}^{p=2} + \overbrace{ADCB \dots YZ}^{p=2} - \overbrace{ACBD \dots YZ}^{p=2} + \dots \\
 &\quad + \overbrace{ABCD \dots :YZ:}^{p=1} - \overbrace{ACBD \dots :YZ:}^{p=2} + \overbrace{ADBC \dots :YZ:}^{p=2} + \dots \\
 &\quad \dots \\
 &\quad + \overbrace{AB :CD \dots YZ:}^{p=1} - \overbrace{AC :BD \dots YZ:}^{p=2} + \overbrace{AD :BC \dots YZ:}^{p=2} + \dots \\
 &\quad + :ABCD \dots YZ: \quad p=0.
 \end{aligned}$$

Contractions $p = n/2$

Wick's theorem

	# Contractions
$ABCD \dots YZ = \overline{AB} \overline{CD} \dots \overline{YZ} - \overline{AC} \overline{BD} \dots \overline{YZ} + \overline{AD} \overline{BC} \dots \overline{YZ} + \dots$	$p = n/2$
$+ \overline{AB} \overline{CD} \dots :YZ: - \overline{AC} \overline{BD} \dots :YZ: + \overline{AD} \overline{BC} \dots :YZ: + \dots$	$p = \frac{n}{2} - 1$
$\dots$	
$+ \overline{AB} :CD \dots YZ: - \overline{AC} :BD \dots YZ: + \overline{AD} :BC \dots YZ: + \dots$	$p = 1$
$+ :ABCD \dots YZ:$	$p = 0.$

Notes

- This is an operator theorem, l.h.s. and r.h.s. are operators!
- This relies on a reference vacuum.
- The true value becomes clear when calculating expectation values.

Remember $\langle \Phi | :A: | \Phi \rangle = 0$, so

$$\begin{aligned} \langle \Phi | ABCD \dots YZ | \Phi \rangle &= \overbrace{AB} \overbrace{CD} \dots \overbrace{YZ} - \overbrace{AC} \overbrace{BD} \dots \overbrace{YZ} + \overbrace{AD} \overbrace{BC} \dots \overbrace{YZ} \\ &= \text{sum of all possible combinations of} \\ &\quad \text{contractions involving all operators} \end{aligned}$$

Wick's theorem: examples

Example 1:

$$\hat{a}_\alpha^\dagger \hat{a}_\beta = \overline{\hat{a}_\alpha^\dagger \hat{a}_\beta} + :a_\alpha^\dagger a_\beta: .$$

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Example 2:

$$\begin{aligned} a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma &= \overbrace{a_\alpha^\dagger a_\beta^\dagger} a_\delta a_\gamma - \overbrace{a_\alpha^\dagger a_\delta} \overbrace{a_\beta^\dagger a_\gamma} + \overbrace{a_\alpha^\dagger a_\gamma} \overbrace{a_\beta^\dagger a_\delta} \\ &+ \overbrace{a_\alpha^\dagger a_\beta^\dagger} :a_\delta a_\gamma: - \overbrace{a_\alpha^\dagger a_\delta} :a_\beta^\dagger a_\gamma: + \overbrace{a_\alpha^\dagger a_\gamma} :a_\beta^\dagger a_\delta: \\ &+ \overbrace{a_\beta^\dagger a_\delta} :a_\alpha^\dagger a_\gamma: - \overbrace{a_\beta^\dagger a_\gamma} :a_\alpha^\dagger a_\delta: + \overbrace{a_\delta a_\gamma} :a_\alpha^\dagger a_\beta^\dagger: \\ &+ :a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma: \end{aligned}$$

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Q: the terms in red vanish; why?

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Q: the terms in red vanish; why?

This equation does not depend on the choice of vacuum and operators.
...but the normal products and contractions do!

Wick's theorem: the nuclear Hamiltonian

Starting from the nuclear Hamiltonian as expressed in a basis $\mathcal{B}_1 = \{a_\alpha^\dagger\}$

$$H = \sum_{\alpha\beta} t_{\alpha\beta} a_\alpha^\dagger a_\beta + \frac{1}{2} \sum_{\alpha\beta\gamma\delta} v_{\alpha\beta\gamma\delta} a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma$$

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Wick's theorem for a Slater determinant in **another** basis $\mathcal{B}'_1 = \{b_\lambda^\dagger\}$.

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The kinetic part is easy

$$\begin{aligned} H_{\text{kin.}} &= \sum_{\alpha\beta} t_{\alpha\beta} \left(\overline{a_\alpha^\dagger} a_\beta + :a_\alpha^\dagger a_\beta: \right) \\ &= \sum_{\alpha\beta} t_{\alpha\beta} \rho_{\beta\alpha} + \sum_{\alpha\beta} t_{\alpha\beta} :a_\alpha^\dagger a_\beta: . \end{aligned}$$

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Vacuum expectation value $\langle \Phi | H_{\text{kin.}} | \Phi \rangle$ ↑

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Vacuum expectation value $\langle \Phi | H_{\text{kin.}} | \Phi \rangle$ ↑

↑
Normal ordered one-body operator
particle hole excitations of the vacuum

Wick's theorem: the nuclear Hamiltonian

The two-body part is more difficult

$$H_{2b} = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} v_{\alpha\beta\gamma\delta} \left[\overbrace{a_{\alpha}^{\dagger} a_{\gamma}} a_{\beta}^{\dagger} a_{\delta} - \overbrace{a_{\alpha}^{\dagger} a_{\delta}} a_{\beta}^{\dagger} a_{\gamma} - \overbrace{a_{\alpha}^{\dagger} a_{\delta}} : a_{\beta}^{\dagger} a_{\gamma} : \right. \\ \left. + \overbrace{a_{\alpha}^{\dagger} a_{\gamma}} : a_{\beta}^{\dagger} a_{\delta} : + \overbrace{a_{\beta}^{\dagger} a_{\delta}} : a_{\alpha}^{\dagger} a_{\gamma} : - \overbrace{a_{\beta}^{\dagger} a_{\gamma}} : a_{\alpha}^{\dagger} a_{\delta} : + : a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma} : \right]$$

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Focussing on the red part

$$\begin{aligned} \frac{1}{2} \sum_{\alpha\beta\gamma\delta} v_{\alpha\beta\gamma\delta} \left[\overline{a_{\alpha}^{\dagger} a_{\gamma}} \overline{a_{\beta}^{\dagger} a_{\delta}} - \overline{a_{\alpha}^{\dagger} a_{\delta}} \overline{a_{\beta}^{\dagger} a_{\gamma}} \right] &= \frac{1}{2} \sum_{\alpha\beta\gamma\delta} v_{\alpha\beta\gamma\delta} [\rho_{\gamma\alpha} \rho_{\delta\beta} - \rho_{\delta\alpha} \rho_{\gamma\beta}] \\ &= \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} \rho_{\gamma\alpha} \rho_{\delta\beta} \end{aligned}$$

$\bar{v}_{\alpha\beta\gamma\delta} = v_{\alpha\beta\gamma\delta} - v_{\alpha\beta\delta\gamma}$

Wick's theorem: the nuclear Hamiltonian

You can keep doing similar tricks to end up with

$$H_{2b} = \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} \left[\frac{1}{2} \rho_{\gamma\alpha} \rho_{\delta\beta} + \rho_{\gamma\alpha} :a_{\beta}^{\dagger} \hat{a}_{\delta}: + \frac{1}{4} :a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}: \right]$$

Wick's theorem: the nuclear Hamiltonian

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$$H_{2b} = \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} \left[\underbrace{\frac{1}{2} \rho_{\gamma\alpha} \rho_{\delta\beta}}_{\text{Vacuum expectation value}} + \underbrace{\rho_{\gamma\alpha} :a_{\beta}^{\dagger} \hat{a}_{\delta}:}_{\text{One-body operator}} + \underbrace{\frac{1}{4} :a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}:}_{\text{Two-body operator}} \right]$$

Wick's theorem: the nuclear Hamiltonian

Defining

$$h_{\alpha\beta}[\rho] = t_{\alpha\beta} + \sum_{\gamma\delta} \bar{v}_{\alpha\gamma\beta\delta} \rho_{\delta\gamma}.$$

Wick's theorem: the nuclear Hamiltonian

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we end up with

$$H = \langle \Phi | H | \Phi \rangle + \sum_{\alpha\beta} h_{\alpha\beta}[\rho] :a_{\alpha}^{\dagger} a_{\beta}: + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} :a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}:$$

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Looking ahead:

- H_1 is the source of all complexity; sometimes called 'residual' interaction.
- Most many-body approaches start by treating H_1 as a perturbation
- Mean-field approximation = variational optimisation of $\langle \Phi | H | \Phi \rangle$.

The extended Wicks theorem

Having treated $ABCD \dots YZ$, what should we do when attempting

$$\langle \alpha\beta | ABCD \dots YZ | \mu\beta \dots \rangle = \langle \alpha\beta | ABCD \dots YZ a_{\mu}^{\dagger} a_{\alpha} | \alpha\beta \dots \rangle$$

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Notice:

- We just expressed $ABCD \dots YZ$ in terms of normal-ordered operators
- $a_{\mu}^{\dagger} a_{\alpha} = \beta_{\mu}^{\dagger} \beta_{\alpha}^{\dagger}$ is already normal-ordered

Reminder: β 's are quasiparticle operators

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Q: why?

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Extending Wicks theorem

When applying Wicks theorem to a string of creation/annihilation operators, you can **omit** all terms containing at least one contraction whose operators both belong to a substring that is already normal-ordered.

Extended Wicks theorem: an example

Let's calculate

particle-hole excitation of a Slater determinant $|\Phi\rangle$

$$\langle \Phi_i^a | :F: | \Phi_j^b \rangle$$

One-body operator $\hat{F} = \sum_{\alpha\beta} f_{\alpha\beta} :a_{\alpha}^{\dagger} a_{\beta}:$
normal-ordered w.r.t. $|\Phi\rangle$

$$\langle \Phi | :a_i^{\dagger} a_a :: a_{\alpha}^{\dagger} a_{\beta} :: a_b^{\dagger} a_j : | \Phi \rangle = \langle \Phi | : \overbrace{a_i^{\dagger} a_a}^{\quad} :: \overbrace{a_{\alpha}^{\dagger} a_{\beta}}^{\quad} :: \overbrace{a_b^{\dagger} a_j}^{\quad} : | \Phi \rangle + \langle \Phi | : \overbrace{a_i^{\dagger} a_a}^{\quad} :: \overbrace{a_{\alpha}^{\dagger} a_{\beta}}^{\quad} :: \overbrace{a_b^{\dagger} a_j}^{\quad} : | \Phi \rangle$$

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$$\langle \Phi | :a_i^{\dagger} a_a :: a_{\alpha}^{\dagger} a_{\beta} :: a_b^{\dagger} a_j : | \Phi \rangle = \langle \Phi | : \overbrace{a_i^{\dagger} a_a}^{\quad} :: \overbrace{a_{\alpha}^{\dagger} a_{\beta}}^{\quad} :: \overbrace{a_b^{\dagger} a_j}^{\quad} : | \Phi \rangle + \langle \Phi | : \overbrace{a_i^{\dagger} a_a}^{\quad} :: \overbrace{a_{\alpha}^{\dagger} a_{\beta}}^{\quad} :: \overbrace{a_b^{\dagger} a_j}^{\quad} : | \Phi \rangle$$

Why only two terms?

Extended Wicks theorem: an example

Let's calculate

particle-hole excitation of a Slater determinant $|\Phi\rangle$

$$\langle \Phi_i^a | :F: | \Phi_j^b \rangle$$

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- All contractions that don't conserve particle number vanish

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- All contractions within a normal ordered product vanish

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Why only two terms?

- All contractions that don't conserve particle number vanish
- All contractions within a normal ordered product vanish
- Once you pick the first contraction (e.g. the partner of $a_i^{\dagger}$), the rest is fixed

Extended Wicks theorem: an example

The job to evaluate the six contractions remains to be done.

- the vacuum is $|\Phi\rangle = \prod_{\alpha=1}^N a_{\alpha}^{\dagger} |0\rangle$
- the relevant annihilation operators are quasi-particle ones (i.e. β_{λ} 's)
- we are interested in p-h excitations $|\Phi_i^a\rangle$ and $|\Phi_j^b\rangle$.
- ...meaning that i, j (a, b) are occupied(empty) in $|\Phi\rangle$

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$$\overbrace{a_i^{\dagger}a_{\beta}}^{\quad} = \rho_{\beta i} = \delta_{\beta i}$$

$$\overbrace{a_a a_b^{\dagger}}^{\quad} = \delta_{ab} - \rho_{ab} = \delta_{ab}$$

$$\overbrace{a_{\alpha}^{\dagger}a_j}^{\quad} = \rho_{j\alpha} = \delta_{j\alpha}$$

$$\overbrace{a_i^{\dagger}a_j}^{\quad} = \rho_{ji} = \delta_{ij}$$

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Don't forget about the signs!

$$\overbrace{a_i^{\dagger} a_{\beta}}^{\text{contract}} = \rho_{\beta i} = \delta_{\beta i}$$

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Reminder

Exercise session 1 will be available by the end of the week.